Insurance Collusion and Imperfect Competition when Insurers Increase Risk*

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Abstract

We study insurers' behavior under collusion and imperfect competition when they can lobby to increase the probability of accidents (e.g., by postponing the introduction of autonomous vehicles or opposing Telematics). We identify a set of cases in which elimination of competition between insurers discourages insurers from lobbying to increase risk. This occurs particularly when initial risk in the industry is high, the marginal cost of lobbying is low, and when consumers' risk aversion is less decreasing in wealth. We also show that insurers would not increase negligible risks if their capability of lobbying to increase risk is sufficiently limited.

Keywords: Insurance, collusion, insurance duopoly, demand for insurance, loading factor, risk, risk aversion, loss modification by insurers.

JEL Classification Numbers: L11, L12, L13, L41, L43, L51, K21, K23

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1 Introduction

The theory of insurance suggests insurers incentivize their insureds to take optimal care thus reducing total risk (Shavell 1979; Shavell 1982). The vast law and economics literature with this focus also includes, for example, Dari-Mattiacci and Parisi (2003), Ben-Shahar and Logue (2012), and Baker and Farrish (2005). Of course, once insurance contracts are signed, insurers wish to mitigate their payouts. However, given that insurers can affect risk before insurance contracts are signed, they may wish to raise risk if this increases their profitability from insurance contracts. Our model focuses on the latter incentives.

Indeed, in various occasions, insurance companies have lobbied to increase the risk from accidents.[1] For example, the National Association of Mutual Insurance Companies ("NAMIC"), whose members represent 39 percent of the U.S. property/casualty insurance market, lobbied in favor of creating obstacles to the introduction of autonomous vehicles.[2] Autonomous vehicles are known to involve a significantly smaller probability of accidents (Shepardson (2019)). Yet, with the introduction of autonomous vehicles, the auto insurance industry would shrink an estimated 60% by 2050 (Hammond (2018)). Indeed, in May 2018, Berkshire Hathaway Chairman and CEO Warren Buffett said in a shareholder event that: "Net it will be bad for the auto insurance industry over time if autonomous cars become a big part of the fleet ... " (Berkshire Hathaway (2018)). As another example, the above-mentioned insurers' association NAMIC has opposed a regulatory requirement to have drivers install Telematics (devices attached to the car or smartphone that track vehicle movements to assess driving skills) (NAMIC (2022). While NAMIC argues that such a requirement might harm drivers' privacy or competition among insurers, it is not clear why insurers would pick that as their fight. Under our framework, in contrast, the answer seems clear: such a requirement would induce drivers to take more precautions thereby reducing overall risk. But as we will show, in-

[1] In 2024, the insurance industry ranked as the third-largest in lobbying expenditures (https://www.statista.com/statistics/257364/top-lobbying-industries-in-the-us/). Some of the lobbying examples below are also cited in Avraham and Porat (2023).

[2] In 2018, NAMIC advocated for creating various obstacles to GM's initiative seeking regulatory changes that would enable it to market its autonomous vehicles (Shepardson 2019). Furthermore, NAMIC has advocated for subjecting autonomous vehicles to more stringent requirements, of third-party validation of design and testing, while ordinary vehicles are under a more lenient self-certification regulatory standard. See https://www.namic.org/pdf/publicpolicy/1805automatedDriving.pdf.

surers are not necessarily interested in such risk reduction. It was also reported that the insurance industry has lobbied against legislative measures fighting high prices of drugs.[3] Another example involves insurers' lobbying efforts in the 1970's for airbags and against seat belts. Given the regulatory framework at the time, in which installing only one of these precautions would suffice, insurers preferred airbags. In later studies, however, the Department of Transportation and researchers have shown that total fatalities and serious injuries would be substantially larger with airbags than with seat belts (DOT 1984; Kneuper and Yandle 1996).

Insurers' lobbying efforts have also allegedly targeted life-saving and risk-reducing genetic tests for the prospects of severe illness. For example, insurers were alleged to create unjustified restrictions for genetic testing (Genomeweb (2015)). Also, they have collectively lobbied against laws forbidding them to discriminate among consumers based on the results of such genetic tests (Rothstein 2008; Otlowski 2002). Had they been allowed to discriminate, many consumers might have preferred to forgo such tests in order not to lose coverage. Of course, insurers' primary motivation here could have been to fit premiums to risk. Nevertheless, it is reasonable to assume insurers were fully aware of the strong disincentives to take such tests that discrimination would cause, thereby substantially increasing the overall risk of severe avoidable illness. Another indirect way insurers can allegedly increase risk is by investing heavily in risky industries, such as fast food and coal (Avraham and Porat 2023).

If insurers use their ability to increase risk too often, or too overtly, such behavior may be prohibitively costly due to negative public opinion. Yet, given our findings that increasing risk can be profitable to insurers, we should expect insurers to act on this incentive, albeit in more subtle ways, perhaps especially when they can provide alternative explanations for such activities, or lobby indirectly, via other entities, as demonstrated in the above-mentioned examples. Moreover, insurers' profitability from increasing risk can shed light on various debates that insurers take part in. For example, NAMIC, the above-mentioned insurers' association, has also advocated against state laws promoted by the National Association of Insurance Commissioners (NAIC) restricting

[3]See, e.g., Biryukov (2022). For example, the insurance industry is represented in the Council for Affordable Healthcare Coverage, a lobbying group that consistently opposes price controls for drugs or importation of drugs that would reduce their costs (e.g., Wanneh and Karins (2025) and Cohrs (2019). Insurers are also active members of the Chamber of Commerce, which successfully opposed 2019 legislation facilitating importation of drugs (Chamber of Commerce (2019)).

the use of inferior non-original auto parts based, inter alia, on safety concerns.[4] Here too, the insurance lobby may have it right, and such regulations may be the result of unwarranted lobbying efforts by original equipment manufacturers. Our results imply, however, that it could be the other way around: Insurers may advocate for unlimited use of inferior non-original auto parts despite the fact that, or even because, such use increases risk.

These examples of insurers lobbying to increase risk also illustrate variation in the reputational marginal costs of lobbying. Opposing autonomous vehicles, for instance, offers few benign justifications, so the reputational cost of such lobbying is likely high. A similar logic applies to lobbying against drug price controls, though insurers mitigate reputational risk by channeling their support through third parties rather than engaging directly. By contrast, lobbying in favor of inferior spare parts may carry lower reputational costs, as it can be framed in terms of reducing payouts or other efficiency concerns. Indeed, we show that the marginal cost of lobbying—which encompasses reputational considerations, among other factors—is an important parameter shaping our results.

The motivating examples above suggest, at the very least, that insurers are able and willing to lobby so as to increase the risk from accidents. This introduces three questions that we address in this paper. First, in a framework like ours, where the premium is proportional to expected harm, what are the exact mechanisms that drive insurers' incentives to lobby for increased risk of accidents? In particular, since increased risk not only increases demand but also the insurer's price-cost margin, might insurers increase risk in way that reduces demand? Second, can we identify circumstances where these incentives to increase risk are negligible, or at least mitigated? Finally, taking insurers' incentives to increase risk as given, could they justify, at least in some circumstances, the often criticized exemption from Federal antitrust laws that insurers enjoy in the United States? In particular, if insurers collude with each other to eliminate competition among them, are they less likely to lobby to increase risk?

In our model, in the first stage, insurers can lobby to increase the probability of accidents (e.g., by lobbying to create obstacles for autonomous vehicles) above the prevailing probability. In the second stage, insurers set their quantities,[5] and in the third stage

[4] See NAMIC (2022a); Bratton and Avila (1999); NAIC (2000).

[5] A quantity-setting framework can be justified by the fact that insurers first decide on the level of capital they dedicate to a particular insurance market (i.e., the level of capacity they have) and only then

each consumer decides on her level of insurance coverage.

We find that when the premium is proportional to expected harm, insurers may wish to increase the probability of accidents for two distinct reasons: first, to increase demand for insurance, and second, to increase insurers' price-cost margin given demand. In particular, the premium the consumer pays is equal to her coverage times qPH, where $q \geq 1$ is the "loading factor", which possibly allows the insurer to make a positive price-cost margin, P is the probability of an accident and H is the level of harm from an accident.[6] Such an insurer's profits are equal to the insurance coverage chosen by consumers times $(q-1)PH$. When the insurer increases P (e.g., by lobbying against autonomous vehicles), it can raise risk-averse consumers' elected coverage given q (a "demand-increasing effect"). Yet, at the same time, above and beyond the demand-increasing effect, raising P increases the insurer's price-cost margin given any $q > 1$. We call this second effect the "market power utilization" effect enabled by the increase in risk, to distinguish it from the demand-increasing effect. As we show, increasing risk can be a profitable way to utilize the insurer's market power.[7] Raising the probability of accidents always serves as a tool for the utilization of market power. Demand, however, does not always increase with the probability of accidents. Yet, because raising the probability of accidents can also be motivated by utilization of market power, and not only by the mere increase in demand, we find that an insurer may raise the probability of accidents in a way that reduces demand.

We show, however, that where the risk of accidents in the relevant activity is negligible in the first place, insurers would not benefit from increasing risk if their capability of

decide on premiums (see Kreps and Scheinkman (1983), who generally show this to be the case under concave demand functions, an efficient rationing rule and symmetric convex cost functions, including zero costs. Wu et al (2012) extend their result to a concave industrial revenue function and convex asymmetric costs). Notably, the common practice of insurance regulators in the U.S. is to evaluate whether insurers had allocated sufficient capital to cover the insured risks (see, e.g., guidelines of the National Association of Insurance Commissioners at https://content.naic.org/sites/default/files/inline-files/Att%20C%20Combined.pdf). There are various previous studies modeling insurance oligopoly as Cournot competition, including Wang et al (2003); Froot and O'Connell (2008); Gao et al (2016); Guo et al (2022); Barreto et al (2023); Liu et al (2025). Cheng and Powers (2008) assume the insurer strategically makes quantity offers to independent underwriters. Gron (1994) finds evidence of capacity constraints in insurance markets.

[6] This framework follows a long line of important papers in insurance economics, starting with Smith (1968); Mossin (1968); Doherty (1975); Hoy and Robson (1981); Doherty and Schlesinger (1983); Schlesinger and Schulenburg (1987); Briys et al (1989); Doherty and Schlezinger (1990); Aase (2007) and Hofmann et al (2009).

[7] We expect this feature to carry over to any case in which an increase in the probability of accidents directly enhances the insurer's price-cost margin.

doing so is too limited. Also, the higher is the initial probability of an accident before the insurer's intervention, the smaller the increase in risk that the insurer lobbies for.

We then study whether insurers' incentive to increase the probability of accidents can help justify the McCarran-Ferguson Act, which exempts the insurance industry from most of the Federal antitrust prohibitions (including price fixing), in the U.S.[8]. Whether price fixing among insurers is allowed in the U.S. differs from state to state, depending on whether or not the state applies its own antitrust prohibitions to the insurance market and if so, to what extent. Accordingly, the case of a fully collusive industry corresponds to cases in which insurers are exempt from antitrust laws and regulation of the local insurance market delegates the task of setting rates to bureaus controlled by the insurers.[9] Full blown collusion also corresponds to insurance markets in which tacit collusion is sustainable. Even a jurisdiction that does not explicitly delegate rate-setting to the insurance market could often encourage information-exchanges among insurers that facilitate tacit collusion.[10] Furthermore, it has been reported that several U.S. states' insurance regulators do not vigorously enforce state antitrust laws, and some state antitrust laws even exempt the insurance industry altogether (Powrie 2010; U.S. Senate 2006, 2007).

Some U.S states are characterized, instead, by (imperfect) competition: while various insurance markets in these states are concentrated,[11] rate regulation is often non-existent, lax, or redundant, and collusion is either blocked by antitrust laws or is not sustainable by the industry. For example, it was reported that rates are determined by free market competition in most insurance markets of U.S. states such as Illinois and Wyoming and in commercial property insurance markets in many other U.S. states such as Arizona,

[8]Formally, the exemption from Federal antitrust law applies only if the state regulates the insurance market, but virtually all states in the U.S. do so in a way that qualifies for such an exemption (see, e.g., Powrie (2010); Schwarcz (2018a) and Anderson (1983)). Although such a regulatory framework exists in practically all states, many of these states impose lenient regulation, so that insurers have considerable leeway despite de jure regulation (see, e.g., Anderson (1983); Areeda and Hovenkamp (2020)). One exception is the health insurance industry, in which this exemption has been limited by the 2021 Competitive Health Insurance Reform Act (CHIRA).

[9]Such is the case, for example, in North Carolina (Schwarcz 2018b; Lehmann 2020; Theodorou 2024). Likewise, many insurance markets in Japan have their rates affected by industry rating organizations (e.g., Dionne (2002); https://www.giroj.or.jp/english/about_1.html and https://www.sonpo.or.jp/en/links/004.html).

[10]The types of exempt information exchanges that often exist in insurance markets include, for example, sharing of projected trends in pricing or costs, and this may facilitate full-blown tacit collusion (Malone (2021)).

[11]Many insurance markets are indeed concentrated (e.g., Austin and Hungerford 2010; Bajtelsmit and Bouzouita 1998; Nissan and Caveny 2000; Cummins et al 1972).

Colorado, Georgia, Kansas, Louisiana, Nebraska, New Mexico, Pennsylvania, Rhode Island, South Dakota and Virginia (D'arcy 2002; PCI 2010; Schwarcz 2018b; Theodorou 2024). Dionne (2002) documents that this is also the case in most segments of auto insurance in Quebec and France.

We identify a set of cases in which insurers' incentive to increase the probability of accidents is larger under imperfect competition than when insurers collude to eliminate competition between them. This occurs particularly when initial risk in the industry (before the insurers' intervention) and the corresponding harm from accidents, are high, and the marginal cost of lobbying is low. It is also more likely to occur the less decreasing in wealth is consumers' risk aversion, making the demand for insurance more steep. The reason is strategic: when insurer A lobbies to increase the probability of accidents, this can induce insurer B to compete less aggressively, and this benefits insurer A. Put differently, raising the probability of accidents under these circumstances enables insurer A to more profitably utilize its market power, but at the same time reduces demand. However, insurer A internalizes all of its profits from utilizing its market power while it externalizes part of the decrease in demand on its rival, insurer B, and at the same time induces more collusive behavior on the part of both firms. This effect does not exist in the case of a collusive industry, in which there is no strategic effect. If this strategic effect is sufficiently strong, it can cause a duopolist to raise the probability of accidents more than would a collusive industry.

Under such circumstances, although collusion increases premiums, it also reduces the probability of accidents, because a collusive industry has weaker incentives to lobby to increase the probability of accidents. This can at least partly justify the controversial exemption of the insurance industry from Federal Antitrust laws.

We contribute to Schlesinger and Venezian (1986), who study an insurer that is able to *reduce* the probability of accidents. Although they focus on risk reduction, they also discuss, in passing, the case where the insurer can raise the probability of accidents and mention that "[a]lthough the insurer may wish to increase the probability of a loss-perhaps by lobbying Congress to block some proposed safety legislation, such action is likely to meet with resistance from the individual ... as well as from insurance regulators." Schlesinger and Venezian (1990) extend Schlesinger and Venezian (1986) to assume that insurers can influence how harm is distributed and show that a monopoly insurer aims

to reduce the size of small losses while keeping large losses significant. Schlesinger and Venezian (1986; 1990) also find that monopoly could benefit consumers compared with (perfect) competition, because with perfect competition, insurers lack the incentive to reduce risks. In contrast, we find that monopoly (or, equivalently, collusion) could benefit consumers compared with (imperfect) competition because under imperfect competition insurers may *increase* risk more than under monopoly or collusion. Hofmann et al (2009) derive a downward sloping demand function for an insurer when the premium is proportional to expected harm, while assuming that the risk from accidents is exogenously given. Our model relies on theirs but assumes instead that insurers can raise the probability of accidents. Ben-Shahar and Logue (2012) considered the claim that, without risk, insurers would be out of business and provided reasons why they are nevertheless incentivized to reduce risk. These reasons do not arise in our framework. In particular, we abstract here from the possibility highlighted by Ben-Shahar and Logue (2012) that an insurer would gain a competitive advantage by introducing innovative ways to reduce its customers' risks. Hinloopen and Soetevent (2020) use a lab experiment to support their theory that an insurer with buyer power may prefer that suppliers charge more (at least from uninsured consumers) to keep harm high, even under perfect competition, so as to raise demand thereby preserving the market for insurance. Soetevent and Zhou (2015) examine several parametric examples of utility functions in which insurers may want to increase the probability of accidents to nearly $\frac{1}{2}$ and extend their result to consumers with loss-aversion.

Unlike these papers, in our framework, the premium is proportional to expected harm, while in their frameworks the premium is not. Hence in their frameworks, an increase in risk can only increase demand, i.e., the consumer's willingness to pay for full coverage (or the number of consumers buying full coverage). This is while in our framework, an increase in risk does not only raise demand for insurance (the consumer's decision as to her level of (possibly partial) coverage given the loading factor), but also serves as an additional tool to exploit market power, and this drives many of our results. Second, unlike the above-mentioned papers, we assume competition is imperfect, introducing a novel strategic interplay among insurers. This enables us to compare between collusion and imperfect competition and discuss a possible justification for the antitrust exemption of the insurance industry.

Avraham and Gilo (2025) show that when the insurer's rates are set by a regulator, non-cost-based rate regulation can be used to disincentivize the insurer from increasing harm and incentivize it to decrease harm. Avraham and Porat (2023), in an informal legal policy paper, analyze the legal implications of insurers' risk enhancing activities.

We also contribute to the literature analyzing consumers' coverage decisions when the premium is proportional to expected harm, starting with Smith (1968) and Mossin (1968) and developed by many other important papers, including, for example, Doherty (1975); Hoy and Robson (1981); Doherty and Schlesinger (1983); Schlesinger and Schulenburg (1987); Briys et al (1989); Doherty and Schlezinger (1990); and Aase (2007). Our paper also contributes to literature analyzing market failures stemming from insurance, such as Parchomovsky and Siegelman (2022) (studying third party moral hazard).[12]

The rest of the paper is organized as follows. Section 2 includes the model. Section 3 studies insurers' incentives to raise the probability of accidents under collusion and imperfect competition. Section 4 concludes.

2 The model

Consider a risk neutral insurer facing a continuum of risk averse consumers. All consumers face a risk of an accident causing harm of $H > 0$ with probability $0 < P < 1$. Consumers differ in their risk aversion (and thus in their willingness to pay for insurance). In particular, consumers have a continuum of risk aversion levels, $\theta \in [\underline{\theta}, \overline{\theta}]$ that is distributed according to a continuous and differentiable distribution function $F(\theta)$ with density $f(\theta)$. Assume θ is not affected by P. The insurer cannot observe an individual consumer's willingness to pay for insurance but knows the distribution of consumers.[13] Denote the utility function of a consumer of type θ as $U_\theta(A)$ where A is the

[12]There is a small number of papers studying lobbying that affects strategies in oligopoly in other contexts, particularly regarding emissions, such as Ehrhart et al (2008) and André and de Castro (2015).

[13]We abstract from the case where insurers offer menus of contracts aimed to discriminate between buyers based on their willingness to pay. Notably, in practice, discrimination by insurers on the basis of consumers' willingness to pay is strictly forbidden according to U.S. rules, which ban price discrimination that cannot be reasonably explained by differences in expected costs (Theodorou 2024; Shapo 2020). As the National Association of Insurance Commisioners (NAIC), emphasizes "[u]nfair discrimination exists if, after allowing for practical limitations, price differentials fail to reflect equitably the differences in expected losses and expenses." (NAIC (2010)). NAIC repeats this prohibition in its 2024 handbook on rate regulation, NAIC (2024), emphasizing that "[a] rule of thumb is that prices are not unfairly discriminatory when consumers are charged different amounts that are actuarially justified (or justified based on risk/cost)." See also Schwarcz (2018b), Robertson v. Ticor Title Ins. Co. (Court of Appeals

consumer's wealth and $U_\theta(A)$ is three times continuously differentiable, with $U'_\theta(A) > 0$ and $U''_\theta(A) < 0$. The consumer's initial wealth is $W > H$.

In the collusive industry case, the industry sets a loading factor $q \geq 1$. A consumer of type θ, given q, H, and P, elects insurance coverage of $\alpha^*_\theta \in [0,1]$ so as to maximize her expected utility $(1-P)U(W_1^{*\theta}) + PU(W_2^{*\theta})$ (where $W_2^{*\theta}$,$W_1^{*\theta}$ are the consumer's levels of wealth with and without an accident, respectively), and buys at most one insurance policy. Given the loading factor q set by the industry, the consumer pays $\alpha^*_\theta qPH$, so the premium is proportional to the expected harm PH and to the loading factor q. For $q = 1$, the premium is actuarially fair and the consumer pays a premium equal to the expected harm covered, $\alpha^*_\theta PH$. But for any $q > 1$, the premium is supra-competitive and the industry has a positive price-cost margin, equal to $(q-1)\alpha^*_\theta PH$. The level of coverage α^*_θ chosen by the consumer is implicitly defined by her first order condition:

$$-(1-P)qPHU'_\theta(W_1^{*\theta}) + PH(1-qP)U'_\theta(W_2^{*\theta}) = 0 \tag{1}$$

As shown in the online appendix, for any $q > 1$ each consumer elects partial coverage (see Doherty (1975)), and for all positive P and H, for the consumer to buy coverage, q must be smaller than $\frac{1}{P}$. We can use α^*_θ to derive the demand function the insurer faces, which aggregates the insurance coverage chosen by all consumers:

$$x(q) \equiv \int_{\underline{\theta}}^{\overline{\theta}} \alpha^*_\theta f(\theta) d\theta \tag{2}$$

3 Insurers raising the probability of accidents

Assume now that insurers can lobby to raise the probability of accidents, P.

The next lemma determines the effect of P on demand:

Lemma 1. *(The effect of P on aggregate demand): Let $R_a(A) \equiv -\frac{U''(A)}{U'(A)}$ denote the Arrow-Pratt coefficient of absolute risk aversion.*

(i) If for all consumer types $\theta \in [\underline{\theta}, \overline{\theta}]$ $R_a^\theta(A)$ is sufficiently decreasing such that $R_a^\theta(W) - R_a^\theta(W - H(1-\alpha^*_\theta)) < \frac{1-q}{\alpha^*_\theta qH}$ and $R_a^\theta(A)$ decreases in a sufficiently slow pace (i.e., $R_a^{\theta\prime}(A)$ is not too increasing), then there exists a probability of accidents $\overline{P}$ $(0 < \overline{P} < 1)$

of Indiana) 982 N.E.2d 9 (2012).

that maximizes aggregate demand and aggregate demand is single peaked in the probability of accidents.

(ii) If for all consumer types $\theta \in [\underline{\theta}, \overline{\theta}]$ $R_a^\theta(A)$ is not too decreasing such that $R_a^\theta(W_1^{*\theta}) - R_a^\theta(W_2^{*\theta}) > \frac{1-q}{\alpha_\theta^* qH(1-Pq)(1-P)}$ for all $P \in (0,1)$, then demand monotonically decreases with P.

Lemma 1 is based on the fact that, at least for a high enough probability of accidents, demand for insurance decreases in this probability. Intuitively, when the probability of an accident is large enough, the gross premium the consumer has to pay becomes closer to the loss itself, and this lowers the marginal utility from buying insurance. At the extreme, as P approaches 1, there is no risk, so there is no marginal utility from buying insurance. Part (i) of the lemma pins down the condition for demand to be non-monotonic in the probability of accidents. Here, when the probability of an accident is small enough and risk aversion is sufficiently decreasing so that the consumer places enough weight on the case of an accident, the consumer wishes to increase her coverage levels as P increases. This is so until P reaches a threshold, $\overline{P}$, in which demand is maximized, while for any $P > \overline{P}$, demand decreases in P.[14] Part (ii) of the lemma characterizes the converse case, in which risk aversion is not too decreasing. Here, the consumer places more weight on the case of no-accident, where she needs to pay the premium (which is rising in P) while bearing no harm, so that demand monotonically decreases in the probability of accidents.[15]

What is particularly important for our purposes is that in both of the cases characterized by Lemma 1, there exists a range where demand is decreasing in P. Yet, because insurers' profitability from increasing P stems not only from the mere increase in demand but also from market power utilization, they may wish to increase P despite the fact that this decreases demand. That is, they may prefer to use an increase in P as a tool to utilize their market power.

In what follows, for simplicity of exposition, we make the following assumption:

[14] For example, it can be verified that for $U_\theta(A) = \sqrt{A}$, demand is strictly concave in P and $0 < \overline{P} < 1$.

[15] It is easy to see that for constant absolute risk aversion (CARA), part (ii) of the lemma applies, and demand is decreasing with P, since then the left-hand side of the condition in part (ii) is zero, while the right-hand side is negative. Note that at $P = 0$, the consumer is indifferent between buying and not buying insurance, because the premium is zero. Hence, consumers' first order conditions are not defined at $P = 0$. Yet, as P increases slightly above zero, the consumer wishes to buy insurance coverage, and under the condition in part (ii) of Lemma 1, coverage is maximal under such a low P.

Assumption 1: a) demand is single peaked in P (i.e., consumers' utility functions correspond to part (i) of Lemma 1);

b) $p + \overline{\Delta p} > \overline{P}$, where p denotes the initial probability of an accident before the insurers' intervention, and $\overline{\Delta p}$ is the insurers' maximum ability to lobby to increase the probability of accidents. Hence, assumption 1.b means that insurers are capable of increasing the probability of accidents above $\overline{P}$.

c) $p < \overline{P}$, meaning that before the insurers' intervention, the probability of accidents was below the one maximizing demand.

As noted, Assumption 1 helps make the exposition more simple, by using the fact that demand is single-peaked in P and using the parameter $\overline{P}$ that maximizes demand. The main economic forces motivating our analysis remain intact, however, without Assumption 1. In particular, when demand is monotonically decreasing in P, according to the condition in part (ii) of Lemma 1, insurers may increase risk in a way that reduces demand, and a duopolist may increase risk more than a collusive industry, for strategic reasons, in order to relax competition.

3.1 Collusive industry raising the probability of accidents

Consider a collusive industry's choice of how much to raise the probability of accidents. Timing is as follows: In the first stage, the industry can take measures to raise P. In the second stage, the industry sets the loading factor q, and in the third stage all consumers simultaneously set their coverage levels α_θ^*. We denote by a small p the natural probability of an accident before the insurers' intervention and by $\Delta p \geq 0$ the increase in this probability achieved by the industry. Thus, capital P, the overall probability of accidents, equals $p+\Delta p$. We assume that the industry's ability to increase the probability of accidents is capped, by $\overline{\Delta p}$. For example, there is a limit to the increase in accident probability caused by creating obstacles to autonomous vehicles. The cost of raising the probability of accidents is $K\Delta p$ where $K > 0$. The collusive industry's profit is:

$$\Pi(q, p + \Delta p) \equiv (q - 1)H \cdot (p + \Delta p)x(q, p + \Delta p) - K\Delta p \tag{3}$$

Where $x(q, p+\Delta p)$ denotes demand as a function of q and $p+\Delta p$. Assume marginal

revenue is decreasing in q and in Δp. Taking the derivative of this profit with respect to Δp and using the envelope theorem we get:

$$\frac{\partial \Pi(q_m^*, p+\Delta p)}{\partial \Delta p} = (q_m^* - 1)Hx(q_m^*, p+\Delta p) + (q_m^* - 1)H \cdot (p+\Delta p)\frac{\partial x(q_m^*, p+\Delta p)}{\partial \Delta p} - K \quad (4)$$

Where q_m^* denotes the industry's optimal choice of q. The first term in (4) is positive and reflects the fact that any increase in Δp enables the industry to utilize its market power. The second term has the same sign as $\frac{\partial x(q_m^*, p+\Delta p)}{\partial \Delta p}$. Recall that under the condition in part (i) of Lemma 1, there exists an accident probability $0 < \overline{P} < 1$ that maximizes demand, implying that this second term could be negative. Under the condition in part (ii) of Lemma 1, this term is negative for all P. If (4) is negative even for $\Delta p = 0$, it follows that the initial probability of accidents p is so large that the industry would rather reduce this probability than use its ability to increase it. In this case, in our framework, which focuses only on lobbying to increase P, $\Delta p_m^* = 0$, where Δp_m^* is the level of Δp chosen by the industry. Such cases could explain why insurers may want to reduce the probability of accidents, when the natural risk in the insured activity is too large from the insurers' perspective.[16] Conversely, if the derivative in (4) is positive when $\Delta p = \overline{\Delta p}$ this implies that $\Delta p_m^* = \overline{\Delta p}$. This corresponds to an industry in which the natural level of p, combined with the insurers' maximum ability to increase it, by $\overline{\Delta p}$, still involves a lower probability of accidents than what the industry would have striven for but for the limit on its ability to increase P. In all other cases, the industry will elect an interior solution, $0 < \Delta p_m^* < \overline{\Delta p}$.

The insurance industry's optimization problem in (4) implies that the larger the initial probability of accidents that prevailed before the industry's intervention, the smaller the increase in this probability that the industry lobbies for:

Corollary 1. *If the marginal cost of lobbying is constant, then in any interior solution, any increase in the initial probability of accidents p is accompanied by a one-to-one reduction of the increase in the probability of accidents the industry lobbies for Δp_m^*.*

The intuition for the result in corollary 1 is that the industry's maximization problems both when choosing the increase in the probability of accidents and when choosing the

[16] Such loss reduction is indeed focused upon by Schlesinger and Venezian (1986, 1990).

loading factor depend solely on the sum of p and Δp, i.e. the total probability of accidents after the industry's intervention.

We next examine whether the industry wishes to lobby to increase risk when the initial risk is negligible. We address this in the following remark.

Remark. **Raising the risk of riskless activities**

Another implication of a collusive industry's choice of Δp concerns its behavior with regard to an activity that, absent the industry's intervention, involves negligible risk. Would the industry want to lobby to raise the probability of accidents in such cases in order to create demand for insurance of this activity? The next proposition shows that, if the industry's ability to raise the probability of accidents is limited enough, the answer is no:

Proposition 1. *(Insurers would not want to raise the probability of accidents from zero when their capability of increasing risk is sufficiently limited): If $p \to 0$ and $\overline{\Delta p}$ is small enough, the collusive industry chooses $\Delta p = 0$.*

The intuition for Proposition 1 is that when the risk from accidents approaches zero, the insurers' marginal revenue from increasing the risk from accidents is zero, while the marginal costs of such lobbying efforts are positive. The marginal revenue is zero because the insurers' starting point is disadvantageous, with zero demand and no market power. For lobbying that increases the risk from accidents to yield a profit from this poor starting point, the increase in risk must be sufficiently significant, and this is not possible when the insurers' ability to lobby to increase risk is limited.

Consider cases corresponding to part (i) of Lemma 1, in which demand is non-monotonic and single peaked in P. The next lemma derives the condition for a collusive industry to set $p + \Delta p_m^* > \overline{P}$, that is, the industry prefers to set Δp_m^* so that the probability of accidents exceeds that which maximizes demand. To pursue this, note that in the second stage, the industry optimally sets q_m^*. The industry's first order condition with respect to the choice of q, given the level of Δp it has chosen in the first stage, after simplifying, yields:

$$x(q_m^*, p + \Delta p) = (1 - q_m^*)\frac{\partial x(q_m^*, p + \Delta p)}{\partial q} \tag{5}$$

Substituting (5) into (4) yields:

$$\frac{\partial \Pi(q_m^*, p+\Delta p)}{\partial \Delta p} = -(q_m^*-1)^2 H \frac{\partial x(q_m^*, p+\Delta p)}{\partial q} + (q_m^*-1) H p \frac{\partial x(q_m^*, p+\Delta p)}{\partial \Delta p} - K \quad (6)$$

Lemma 2. *(When does a collusive industry raise P in a way that reduces demand below its maximum level?):*

(i) $p + \Delta p_m^* > \overline{P}$ (the collusive industry increases P in a way that reduces demand below its maximum level) if the ratio of the marginal cost of lobbying to harm $\frac{K}{H}$ is small, i.e.:

$$\frac{K}{H} < -\frac{\partial x(q_m^*, \overline{P})}{\partial q}(q_m^*-1)^2 \quad (7)$$

(ii) The right-hand side of (7) increases the less decreasing in wealth is consumers' risk aversion.

What drives the result in Lemma 2 is that increasing the probability of accidents, P, not only affects demand *per se* but also enables the industry to utilize its market power. This is manifested in the first term of (6). This is why raising P even beyond $\overline{P}$, in a way that reduces demand below its maximum level, may be profitable. The smaller is $\frac{K}{H}$, the more likely is the industry to raise the probability of accidents above the level maximizing demand. Larger harm, H, reinforces the profitability of using the increase in P as a tool to utilize market power. At the same time, the lower the marginal cost of lobbying, K, the less the industry needs to sacrifice in order to raise the probability of accidents in a way that lowers demand.[17] Indeed, what matters is the ratio between the marginal cost of lobbying and harm from accidents. The marginal cost of lobbying, K, could include reputational costs stemming from negative public opinion regarding lobbying to increase risk. These costs depend on how convincing are other excuses for the lobbying efforts. For example, lobbying against autonomous vehicles or drivers' monitoring devices, documented in the introduction, seem to involve high reputational costs, because it is difficult to think of benign motivations for these lobbying efforts. Conversely, lobbying in favor of inferior non-original auto parts may involve lower reputational costs, since there exists a more benign explanation (cost savings) for such lobbying efforts. Similarly, cases in which lobbying efforts are carried out indirectly

[17] Note that as the marginal cost of lobbying approaches zero, condition (7) is always met.

through third parties—for instance, in opposing price controls and drug importation, as noted in the introduction—correspond to situations with low marginal lobbying costs. The example mentioned in the introduction regarding lobbying against genetic tests corresponds to large harm (H), since genetic tests reduce the prospects of severe illness. As noted, given the marginal cost of lobbying, K, a large H supports the condition for a demand-reducing increase in the probability of accidents. Although the supra-competitive premium q itself can be used to exert market power, raising P (even in a way that reduces demand) can be preferred to raising q if the reduction in demand when raising q is strong enough. This is the intuition for the condition in part (i) of the proposition, which implies that, given $\frac{K}{H}$, the larger is $\frac{\partial x(q_m^*, \overline{P})}{\partial q}$ in absolute value (the steeper is demand) the more likely is the industry to increase P beyond $\overline{P}$ in a way that reduces demand.

Part (ii) of the proposition links this result to the question of how increasing or decreasing in wealth is consumers' risk aversion. Given the ratio, $\frac{K}{H}$, the less decreasing is consumers' risk aversion, aggregate demand declines more strongly in the premium q, and this raises the chances that condition (7) holds. Conversely, if consumers' risk aversion is sufficiently decreasing, then the negative effect increasing q has on demand is weaker. In such cases, given $\frac{K}{H}$, the industry is more likely to utilize its market power by increasing q rather than increasing P to levels that reduce demand.

In the next sub-section, we characterize a duopolist's decisions to raise P and compare it to that of a collusive industry.

3.2 Insurance duopoly raising the probability of accidents

Let us now assume that there are two identical insurers, firm 1 and firm 2, who simultaneously set quantities of insurance policies x_1 and x_2 respectively.[18] Consumers see firm 1 and firm 2's insurance policies as perfect substitutes. The inverse demand function is $q(x_1 + x_2)$.[19] Timing is as follows:

[18] As mentioned in the introduction, the quantity-setting framework is equivalent, under reasonable assumptions, to a case where insurers first decide on the level of capital they dedicate to a particular insurance market and only then decide on premiums. This follows the path of many previous studies of insurance oligopoly.

[19] Since the demand function $x(q)$ is one-to-one and strictly downward sloping, there exists an inverse demand function $q(x)$ that identifies the loading factor q as a function of the aggregate quantity $x_1 + x_2$ chosen by the insurers.

1) In the first stage, both insurers simultaneously can take actions to raise the probability of accidents. Denote by Δp_i insurer i's choice of the increase in the probability of accidents ($i = 1, 2$). Since both firms are assumed to be able to influence the level of Δp, we assume the insurer setting the higher level will determine it. Hence $\Delta p = max\{\Delta p_1, \Delta p_2\}$.

2) In the second stage, both insurers simultaneously set quantities x_1 and x_2.

3) In the third stage, the loading factor $q(x_1 + x_2)$ is determined.

Accordingly, if insurer i ($i = 1, 2$) is the one determining Δp in the first stage, its profit is:

$$\Pi_i^c(x_1 + x_2, p + \Delta p) \equiv x_i \cdot (p + \Delta p)H[q(x_1 + x_2, p + \Delta p) - 1] - K\Delta p \tag{8}$$

Assume firm i's marginal revenue is increasing in x_i (given Δp) and decreasing in Δp.

Denote as Δp_c^* the level of Δp maximizing (8). There are two possible sub-game perfect Nash equilibria in the first stage's game. One in which firm 1 increases the probability of accidents by Δp_c^* and firm 2 does nothing to raise risk, and the other in which firm 2 is the one setting Δp_c^* and firm 1 does nothing.

To compare between the behavior of a duopolist and a collusive industry, we can replicate the analysis of Subsection 3.1 by assuming the collusive industry sets the quantity x_m^* instead of the loading factor q_m^*. Hence the industry's profit is $\Pi^m(x) \equiv [q(x, p + \Delta p) - 1]H \cdot (p + \Delta p)x - K\Delta p$. The derivative of this profit with respect to Δp, using the envelope theorem, is:

$$\frac{\partial \Pi^m(x_m^*, p + \Delta p)}{\partial \Delta p} = [q(x_m^*, p + \Delta p) - 1]Hx_m^* + \frac{\partial q(x_m^*, p + \Delta p)}{\partial \Delta p} H \cdot (p + \Delta p)x_m^* - K \tag{9}$$

Accordingly, a quantity-setting collusive industry too chooses Δp_m^* and the results from Subsection 3.1 trivially carry over. We now ask whether a duopoly behaves differently from a collusive industry in its choice of Δp.

In the duopoly case, the full derivative of the duopolist's profit (8) with respect to Δp, using the envelope theorem, is:

$$\frac{d\Pi_i^c(x_1^{*c}+x_2^{*c},p+\Delta p)}{d\Delta p} = x_i^{*c}H\cdot[q(x_1^{*c}+x_2^{*c},p+\Delta p)-1]+x_i^{*c}(p+\Delta p)H\frac{\partial q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial \Delta p}$$
$$+x_i^{*c}(p+\Delta p)H\frac{\partial q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial(x_1+x_2)}\frac{\partial x_j(p+\Delta p)}{\Delta p}-K \tag{10}$$

where x_i^{*c} is firm i's equilibrium quantity, $i=1,2$ and $j\neq i$. The first term in (10) stems from the fact that any increase in the probability of accidents enables the insurer to utilize its market power. The second term represents the direct effect of raising the probability of accidents on demand. By Lemma 1, demand could be decreasing with P. In such cases, the second term in (10) is negative. As we show below, this also affects the sign of the third term in (10), which represents the strategic effect of Δp: if, in response to firm i raising P, firm j contracts output ($\frac{\partial x_j(p+\Delta p)}{\Delta p}<0$), the strategic effect is positive. Conversely, if, in response to firm i raising P, firm j increases output ($\frac{\partial x_j(p+\Delta p)}{\Delta p}>0$), the strategic effect is negative.

When comparing a duopolist's incentive to increase risk to that of a collusive industry, there are four effects to consider that do not necessarily work in the same direction. First, in duopoly, there is a strategic effect (the last term in (10)), as noted above, which is absent in the case of a collusive industry. As we shall see, this strategic effect may be positive, so a duopolist may wish to raise the probability of accidents for strategic reasons: to induce its rival to compete less aggressively, thereby achieving a more collusive outcome. This effect may cause a duopolist to increase risk more than would a collusive industry. Second, there is a "market power utilization effect": the increase in risk improves insurers' ability to utilize their market power, and this is more profitable for a collusive industry, which enjoys higher market power to begin with (for a given $p+\Delta p$, the first term in (4) is always larger than the first term in (10)). This effect works in the opposite direction, possibly inducing a collusive industry to increase risk more than would a duopolist. Third, there is a "quantity effect": since each firm's quantity in duopoly (x_i^{*c}, $i=1,2$) is smaller than the monopoly quantity (x_m^*), the effects of the increased risk on demand and on market power utilization apply to a smaller number of units. Fourth, there is an "aggregate demand" effect: aggregate quantity under duopoly is larger than under collusion, and this can affect the degree to which the increase in risk changes demand. This effect is determined by the sign and size of the cross derivatives

$\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x \partial \Delta p}$, $\frac{\partial^2 q(x_m^*,p+\Delta p)}{\partial x \partial \Delta p}$ in the domain $\Delta p \in (0, \overline{\Delta p})$, which measure how an increase in quantity influences the effect of increasing risk on demand. These cross derivatives also measure how the increase in risk affects the slope of the demand function, i.e., how Δp affects the derivative of demand with respect to quantity. We shall focus on cases in which these cross derivatives are sufficiently small in absolute value. We start from the case where $\frac{K}{H}$, the ratio of the marginal cost of lobbying to harm, is intermediate and the gap between the market power of a collusive industry and a duopolist is sufficiently large:

Proposition 2. *Suppose that the increase in the probability of accidents causes a parallel shift in demand (i.e.,* $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},\overline{P})}{\partial x \partial \Delta p} = \frac{\partial^2 q(x_m^*,\overline{P})}{\partial x \partial \Delta p} = 0$*). If the ratio of the marginal cost of lobbying to harm,* $\frac{K}{H}$*, is intermediate, i.e.:*

$$[q(x_m^*, \overline{P}) - 1]x_m^* > \frac{K}{H} > [q(x_1^{*c} + x_2^{*c}, \overline{P}) - 1]x_i^{*c} \tag{11}$$

then a collusive industry sets $p + \Delta p > \overline{P}$ *and a duopolist sets* $p + \Delta p < \overline{P}$*, so a collusive industry increases the probability of accidents more than does a duopolist (i.e.,* $\Delta p^{*c} < \Delta p_m^*$*).*

As shown in the proof of Proposition 2, the result carries over to cases where the demand shift is not parallel, such that the increase in risk changes the slope of the demand function, but does not steepen this slope too much.

The two sides of the condition in (11) are a collusive industry's and a duopolist's market power (the extent to which the loading factor q exceeds 1) when the probability of accidents is maximal ($P = \overline{P}$), multiplied by their equilibrium quantities. The larger the gap between a collusive industry's market power and that of a duopolist, the more likely is the condition to be met.

Consider now cases where the ratio of the marginal cost of lobbying to harm, $\frac{K}{H}$, is not intermediate pursuant to the condition in Proposition 2 or where the gap between the market power of a collusive industry and a duopolist is not large. Consider, in particular, the case where the ratio $\frac{K}{L}$ is small, so that both a duopolist and a collusive industry increase risk in a way that reduces demand, under the condition $[q(x_m^*, \overline{P}) - 1]x_m^* > [q(x_1^{*c} + x_2^{*c}, \overline{P}) - 1]x_i^{*c} > \frac{K}{H}$, derived in the proof of Corollary 2 below. Here, we identify a set of cases in which a duopoly insurer would lobby to increase risk more than would

a collusive industry. This introduces a possible justification for the antitrust exemption the insurance industry enjoys.

What drives the possibility that a duopolist lobbies to increase risk more than a collusive industry is the strategic effect of raising risk. When a duopolist increases risk in a way that decreases demand, standard comparative statics results of a Cournot duopoly imply that the strategic effect of raising risk can be positive under reasonable and plausible assumptions: the increase in risk by firm i can cause contraction of output by firm j, thereby achieving a more collusive duopoly equilibrium. As shown by Dixit (1986), this occurs if and only if the increase in the probability of accidents decreases the firms' marginal profits. To see this, firm i's marginal profits being decreasing in Δp, taking account of firms' optimal quantity choices x_1^{*c} and x_2^{*c}, means that:

$$\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x_i \partial \Delta p} H \cdot (p+\Delta p) x_i^{*c} + \frac{\partial q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial \Delta p} H \cdot (p+\Delta p) < 0 \tag{12}$$

When demand is decreasing in Δp, the second term in (12) is negative. Hence, if the cross derivative $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x_i \partial \Delta p}$ (for $\Delta p \in (0, \overline{\Delta p})$) in the first term is negative, or not too positive, firm i's marginal profit decreases in risk, and the strategic effect is positive.[20]

Another force driving the possibility that a duopolist lobbies to raise risk more than would a collusive industry is the above-mentioned "quantity effect": when the increase in risk reduces demand, the second terms in (10) and (9) are negative, but the second term in (9) is more pronounced because the number of units supplied by a collusive industry is larger. The only effect that clearly works in the opposite direction, of causing a collusive industry to increase risk more than would a duopolist, is the "market power utilization effect" (the first term in (10) always being smaller than the first term in (9) for a given $p + \Delta p$). As noted, the "aggregate demand effect" (i.e., is the increase in risk more effective in reducing demand in duopoly versus a collusive industry), depends on the cross derivative $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x_i \partial \Delta p}$. We focus below on cases in which this cross derivative is sufficiently small:

Corollary 2. *If the ratio of the marginal cost of lobbying to harm $\frac{K}{H}$ is small, i.e.:*

[20] We relegate the full details of the standard comparative statics to the online appendix.

$$[q(x_m^*, \overline{P}) - 1]x_m^* > [q(x_1^{*c} + x_2^{*c}, \overline{P}) - 1]x_i^{*c} > \frac{K}{H} \tag{13}$$

, and the demand shift is parallel (i.e., $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x \partial \Delta p} = \frac{\partial^2 q(x_m^*,p+\Delta p)}{\partial x \partial \Delta p} = 0$ for $\Delta p \in (0, \overline{\Delta p})$*), then there exists a cutoff* $\widehat{P}$ *such that for any demand-maximizing probability* $\overline{P} > \widehat{P}$*, a duopolist raises the probability of accidents more than a collusive industry (*$\Delta p^{*c} > \Delta p_m^*$*).*

This result extends to cases where the demand shift is not parallel, as long as the increase in risk does not moderate the slope of the demand function to much. As long as the cross derivatives $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x \partial \Delta p}$ and $\frac{\partial^2 q(x_m^*,p+\Delta p)}{\partial x \partial \Delta p}$ are not too positive for $\Delta p \in (0, \overline{\Delta p})$, the strategic effect is guaranteed to be positive (by the condition in (12)), and the aggregate demand effect is negligible. As noted, the only countervailing effect, working in favor of a collusive industry, is the market power utilization effect, which is stronger for a collusive industry. But for parameter values in which the market power utilization effect in the duopoly case is close enough in magnitude to the corresponding effect in the collusive case, the strategic effect and quantity effect, working in the opposite direction, dominate. A sufficient condition for this to occur is when $\overline{P}$ (the probability of accidents maximizing demand) is large enough. Intuitively, a positive strategic effect corresponds to the case where $p + \Delta p > \overline{P}$ (where demand is decreasing in P) so a high $\overline{P}$ implies a high initial probability of accidents p. As the initial probability of accidents before the insurer's intervention (p) increases, the market power effect diminishes, since then market power utilization is significant in the duopoly case as well, and at the same time the (positive) strategic effect strengthens. Also, we have already seen that the smaller the ratio $\frac{K}{H}$, and the less decreasing in wealth is consumers' risk aversion (see also part (ii) of Lemma 2), the more likely we are to be in the region where the increase in risk reduces demand, which is the region in which duopolists are particularly more eager to raise risk for strategic reasons. Hence, higher harm from accidents, H and lower marginal costs of lobbying, K, as well as consumers' risk aversion not being too decreasing in wealth (implying steep demand for insurance), also raise the chances that duopolists increase risk more than a collusive industry.

It is straightforward that the gist of the result in Corollary 2, according to which a duopolist may raise risk more than a collusive industry, extends to cases in which

demand is monotonically decreasing in the probability of accidents (by the condition in part (ii) of Lemma 1). Since demand is decreasing, when duopoly insurers nevertheless raise P to better utilize their market power (i.e., at $\Delta p = 0$, (10) is positive), duopolists may raise P more than would a collusive industry. The driving forces here too are the positive strategic effect and negative quantity effect.

The intuition for a duopolist wanting to increase risk more than would a collusive industry is that if firm i raises the probability of accidents when the strategic effect is positive, it not only utilizes its market power (the first term in (10)) but also induces its rival firm j to contract output (the third term in (10)), leading to a more collusive outcome. Raising P increases firm i's profit via market power utilization, but at the same time reduces demand. Yet, firm i internalizes all of the profits from market power utilization while it externalizes part of the decrease in demand on firm j and at the same time induces more collusive behavior on the part of both firms. This effect is absent in the case of collusion, in which there is no strategic effect.

In the cases identified here, in which duopolists raise the probability of accidents more than a collusive industry, the welfare effects of collusion among insurers become ambiguous – while it raises premiums, it can reduce the probability of accidents.

4 Conclusion

Casual observation suggests that insurers are able and willing to lobby in a way that increases the risk from accidents. We demonstrated the mechanisms that incentivize insurers to do this: when the premium is proportional to expected harm, insurers may lobby to raise the probability of accidents not only to increase demand, but also to better utilize their market power. In fact, the latter motivation for risk-increasing lobbying can induce insurers to increase the probability of accidents even when this reduces demand for insurance. When consumers' risk aversion is not too decreasing in wealth, their demand is more sensitive to premiums. Then, insurers prefer utilizing their market power by lobbying to increase the risk of accidents than by causing premiums to increase. This demand-reducing effect of the boost in risk, in turn, may motivate a duopolist to lobby to increase risk for strategic reasons, in order to relax competition and achieve a more collusive outcome. Such a strategic effect is absent when insurers collude not to compete

with each other. Accordingly, there are cases in which insurers lobby to increase risk more in a duopoly than under collusion. This is more likely to occur when the initial risk of accidents in the market is high and when the marginal cost of lobbying is low. It is also more likely to occur when the gap in the market power of a collusive industry and a duopolist is not large. In such cases, taking insurers' lobbying efforts as given, collusion, while increasing premiums, have the upside of reducing the probability of accidents.

Insurance collusion is permitted in the U.S. under the McCarran-Ferguson Act, particularly in states in which state regulation enables such collusion or facilitates it. North Carolina is a clear example, in which, by state regulations, rates are determined collusively by insurers. Collusion need not be explicit as in North Carolina, and could be tacit, in concentrated local insurance markets. Tacit collusion may be likely given that many states permit practices that may facilitate collusion, such as the exchange of non-public sensitive information among insurers. Such facilitating practices too, if permitted by the states, are perfectly legal, pursuant to the exemption from Federal antitrust law. Since we identify a set of cases where duopolists lobby to increase risk more than would a collusive industry, a counterargument could be advanced against some U.S. policymakers' persistent calls to repeal or narrow the scope of the McCarran–Ferguson Act (see, e.g., U.S. Senate, 2006; 2007), as has occurred in the U.S. health insurance market.

Appendix

Proof of Lemma 1

Totally differentiating (1) with respect to P yields:

$$\begin{aligned}&\tfrac{\partial\alpha^*_\theta(P)}{\partial P}[PHq^2(1-P)U''_\theta(W_1^{*\theta})+(1-qP)^2HU''_\theta(W_2^{*\theta})]\\&=-\alpha^*_\theta q^2H(1-P)U''_\theta(W_1^{*\theta})+\alpha^*_\theta qH(1-qP)U''_\theta(W_2^{*\theta})\\&\qquad -qU'_\theta(W_1^{*\theta})+qU'_\theta(W_2^{*\theta})\end{aligned}\tag{14}$$

The expression in squared brackets in the first line of (14) is negative for all $P\in(0,1)$ since $U''_\theta(A)<0$. At $P\to 1$, the term in the squared brackets on the left hand side of (14) converges to $H(1-q)^2U''_\theta(W_2^{*\theta})<0$ for all $q>1$. Consider now the right hand side of (14) at $P\to 1$. Consumers elect coverage converging to zero, because a consumer's expected utility, $U(W-\alpha^*_\theta qH-(1-\alpha^*_\theta)H)$ is decreasing in α^*_θ. Hence the right hand side converges to $q[U'_\theta(W_2^{*\theta})-U'_\theta(W_1^{*\theta})]>0$ where the inequality follows from the concavity of $U_\theta(A)$. This implies that the consumer's coverage always decreases as P approaches 1.

Let us now examine under what circumstances $\alpha^*_\theta(P)$ increases in P for $P\to 0$. Writing the term in the squared brackets on the left hand side of (14) in terms of the coefficient of absolute risk aversion $R^\theta_a(A)$ and taking account of the fact that, by (1), for $0<P<1$ and $H>0$ $U'_\theta(W_2^{*\theta})=\frac{U'_\theta(W_1^{*\theta})(1-P)q}{1-qP}$, yields that this term equals:

$$-(1-P)qHU'_\theta(W_1^{*\theta})[qPR^\theta_a(W_1^{*\theta})+(1-qP)R^\theta_a(W_2^{*\theta})]\tag{15}$$

The right hand side of (14) similarly yields:

$$\begin{aligned}&(1-P)q^2H\alpha^*_\theta U'_\theta(W_1^{*\theta})[R^\theta_a(W_1^{*\theta})-R^\theta_a(W_2^{*\theta})]\\&\qquad +qU'_\theta(W_1^{*\theta})\tfrac{q-1}{1-qP}\end{aligned}\tag{16}$$

Hence after simplifying we get:

$$\frac{\partial\alpha^*_\theta(P)}{\partial P}=\frac{(1-P)\alpha^*_\theta qH[R^\theta_a(W_1^{*\theta})-R^\theta_a(W_2^{*\theta})]+\frac{q-1}{1-qP}}{-(1-P)H[qPR^\theta_a(W_1^{*\theta})+(1-qP)R^\theta_a(W_2^{*\theta})]}\tag{17}$$

The denominator of (17) is negative for all $P\in(0,1)$. For $P\to 0$, the numerator of (17) converges to:

$$\alpha_\theta^* qH[R_a^\theta(W) - R_a^\theta(W - H(1-\alpha_\theta^*))] + q - 1 \tag{18}$$

If risk aversion sufficiently decreases in wealth so that:

$$R_a^\theta(W) - R_a^\theta(W - H(1-\alpha_\theta^*)) < \frac{1-q}{\alpha_\theta^* qH} \tag{19}$$

then the numerator of (17) is negative, and hence the whole expression is positive, for $P \to 0$. By the preceding analysis, since $\frac{\partial \alpha_\theta^*(P)}{\partial P}$ is continuous for all $P \in (0,1)$, then under the condition in (19), the function $\alpha_\theta^*(P)$ has at least one maximum point in $P \in (0,1)$, where $\frac{\partial \alpha_\theta^*(P)}{\partial P} = 0$. Let $\overline{P}^\theta$ denote the level of $P \in [0,1]$ maximizing α_θ^* for a consumer of type θ.

Aggregate demand is $x(q,p) = \int_{\underline{\theta}}^{\overline{\theta}} \alpha_\theta^*(q,P) f(\theta) d\theta$. Differentiating it by P yields:

$$\frac{\partial \int_{\underline{\theta}}^{\overline{\theta}} \alpha_\theta^*(q,P) f(\theta) d\theta}{\partial P} = \int_{\underline{\theta}}^{\overline{\theta}} \frac{\partial \alpha_\theta^*(q,P)}{\partial P} f(\theta) d\theta$$

Accordingly, by the same reasoning applied to each consumer's individual coverage decision, if the condition in (19) is met there exists at least one maximum of $F(p) \equiv \int_{\underline{\theta}}^{\overline{\theta}} \alpha_\theta^*(q,P) f(\theta) d\theta$ $(P \in (0,1))$ satisfying $\int_{\underline{\theta}}^{\overline{\theta}} \frac{\partial \alpha_\theta^*(q,P)}{\partial P} f(\theta) d\theta = 0$. Let $\overline{P}$ denote the level of P maximizing $F(P)$. The preceding analysis shows such a $\overline{P}$ exists and satisfies $0 < \overline{P} < 1$.

To see why demand is single peaked under the conditions in part (i) of the lemma, recall that the denominator of (17) is always negative. The coefficient multiplying the term in squared brackets in the numerator is decreasing in P. The term in squared brackets in the numerator is negative whenever risk aversion is decreasing in wealth. Hence this term becomes less negative as P increases as long as $R_a^{\theta\prime}(A)$ is not too increasing in A. Accordingly, for decreasing $R_a^\theta(A)$ and a not too increasing $R_a^{\theta\prime}(A)$, the first term in the numerator is negative and becomes less negative as P increases. The second term in the numerator is positive, since $qP < 1$, and increasing in P. This implies that under condition (19), the numerator reaches zero at $\overline{P}^\theta$, and for any $P > \overline{P}^\theta$, by the arguments above, it becomes more and more positive as P increases. Thus for any $P \in (0,1)$, $\alpha_\theta^*(P)$ increases in P until P reaches $\overline{P}^\theta$ and then decreases in P for $P > \overline{P}^\theta$. Since this holds

for any $\theta \in [\underline{\theta}, \overline{\theta}]$, the same sufficient condition applies for $x(q,p) = \int_{\underline{\theta}}^{\overline{\theta}} \alpha_\theta^*(q,P) f(\theta) d\theta$.[21]

Part (ii) of the lemma follows directly from (19).■

Proof of Corollary 1

The Industry's first order condition with respect to Δp is:

$$(q_m^* - 1)Hx(q_m^*, p + \Delta p^*) + (q_m^* - 1)H \cdot (p + \Delta p^*) \frac{\partial x(q_m^*, p + \Delta p^*)}{\partial \Delta p} = K \tag{20}$$

Its first order condition with respect to q is:

$$H \cdot (p + \Delta p^*) x(q_m^*, p + \Delta p^*) + (q_m^* - 1)H \cdot (p + \Delta p^*) \frac{\partial x(q_m^*, p + \Delta p^*)}{\partial \Delta q} = 0 \tag{21}$$

As far as p or Δp are concerned, both (20) and (21) depend only on their sum $p + \Delta p^*$. Hence, after implicitly substituting the solution of (21) for q_m^* back into (20), to maintain both conditions, the right hand sides of which are constant, any increase in p must be accompanied by a one-to-one decrease in Δp^*.■

Proof of Proposition 1

The first expression in the derivative of the industry's profits with respect to Δp (4) converges to zero when $p + \Delta p \to 0$ because $x(q_m^*, p + \Delta p) \to 0$ when $p + \Delta p \to 0$. For $p + \Delta p \to 0$, we can assume $\alpha_\theta^* = 0$ for all $\theta \in [\underline{\theta}, \overline{\theta}]$. Absent risk, premiums are equal to zero regardless of coverage, so all consumers are indifferent to their coverage choice, but if buying coverage involves even a negligible administrative cost, no consumer will buy insurance absent risk. As for the second expression in (4), fully differentiating the consumer's first order condition in (1) with respect to q and rearranging terms yields):

$$\begin{aligned} \tfrac{\partial \alpha_\theta^*(q)}{\partial q} \cdot [q^2 P^2 H^2 (1-P) U_\theta''(W_1^{*\theta}) + (1-qP)^2 P H^2 U_\theta''(W_2^{*\theta})] \\ = PH(1-P)U_\theta'(W_1^{*\theta}) + P^2 H U_\theta'(W_2^{*\theta}) \\ -\alpha_\theta^* P^2 H^2 q (1-P) U_\theta''(W_1^{*\theta}) + \alpha_\theta^* P^2 H^2 (1-qP) U_\theta''(W_2^{*\theta}) \end{aligned} \tag{22}$$

For any consumer of type $\theta \in [\underline{\theta}, \overline{\theta}]$, $\frac{\partial \alpha_\theta^*(q)}{\partial q} \to \frac{U_\theta'(W_1^{*\theta})}{H U_\theta''(W_2^{*\theta})}$ when $p + \Delta p \to 0$. Hence by the insurer's first order condition in (21) $q_m^* \to 1$ when $p + \Delta p \to 0$ because the left hand side of this first order condition approaches zero as $p + \Delta p \to 0$ and in the right hand

[21] Although it suffices for us that demand is single peaked in P, the online appendix derives a sufficient condition for $\alpha_\theta^*(P)$, and therefore for aggregate demand, to be concave in P.

side, $(1-q_m^*)$ is multiplied by an expression approaching $\frac{U'_\theta(W_1^{*\theta})}{HU''_\theta(W_2^{*\theta})}$, which is negative and finite. As $p+\Delta p \to 0$, by (17), for any consumer of type $\theta \in [\underline{\theta}, \overline{\theta}]$, $(p+\Delta p)\frac{\partial \alpha_\theta^*(p+\Delta p)}{\partial \Delta p}$ approaches $\frac{qHU'_\theta(W_1^{*\theta})-HU'_\theta(W_2^{*\theta})}{H^2U''_\theta(W_2^{*\theta})}$. Therefore, the second expression in (4) approaches zero when $p+\Delta p \to 0$ because $(1-q_m^*) \to 0$ and $(p+\Delta p)\frac{\partial x(q_m^*,p+\Delta p)}{\partial \Delta p} \to \frac{qHU'_\theta(W_1^{*\theta})-HU'_\theta(W_2^{*\theta})}{H^2U''_\theta(W_2^{*\theta})}$, which is finite. Therefore, at $p+\Delta p \to 0$, $\frac{\partial \Pi(q_m^*,p+\Delta p)}{\partial \Delta p} \to -K < 0$ and by the continuity of $\frac{\partial \Pi(q_m^*,p+\Delta p)}{\partial \Delta p}$, it remains negative for small enough Δp. It follows that for $\overline{\Delta p}$ small enough and $p \to 0$, $\Delta p_m^* = 0$.■

Proof of Lemma 2

To prove part (i), note that at $p+\Delta p = \overline{P}$, $\frac{\partial x(q,p+\Delta p)}{\partial \Delta p} = 0$, so the second term in (6) vanishes and (6) turns into:

$$\frac{\partial \Pi(q_m^*, \overline{P})}{\partial \Delta p} = -\frac{\partial x(q, \overline{P})}{\partial q} H(q_m^* - 1)^2 - K$$

which is positive if and only if $-\frac{\partial x(q,\overline{P})}{\partial q}(q_m^* - 1)^2 > \frac{K}{H}$.

To prove part (ii), writing the expression of $\frac{\partial \alpha_\theta^*(q)}{\partial q}$ derived in (22) in terms of the coefficient of absolute risk aversion $R_a^\theta(A)$ and utilizing the fact that from the consumer's first order condition in (1), $U'_\theta(W_2^{*\theta}) = \frac{(1-P)qU'_\theta(W_1^{*\theta})}{1-qP}$ we get:

$$\begin{aligned} &\tfrac{\partial \alpha_\theta^*(q)}{\partial q}[-q^2PHR_a^\theta(W_1^{*\theta}) - q(1-qP)HR_a^\theta(W_2^{*\theta})] \\ &\quad = \alpha_\theta^* qPH[R_a^\theta(W_1^{*\theta}) - R_a^\theta(W_2^{*\theta})] + \tfrac{1}{1-qP} \end{aligned} \tag{23}$$

Note that the right hand side of (23) increases with $R_a^\theta(W_1^{*\theta}) - R_a^\theta(W_2^{*\theta})$. This is while the expression in squared brackets in the left hand side of (23) does not depend on whether risk aversion is decreasing or increasing in wealth because both $R_a^\theta(W_1^{*\theta})$ and $R_a^\theta(W_2^{*\theta})$ appear in it with the same sign. Since this holds for all levels of P, and aggregate demand $x(q_m^*, p+\Delta p)$ equals the aggregation of consumers' coverage choices, $\frac{\partial x(q_m^*,p+\Delta p)}{\partial q}$ too grows in absolute value with $R_a^\theta(W_1^{*\theta}) - R_a^\theta(W_2^{*\theta})$. Now going back to condition (7), given the ratio $\frac{K}{H}$, this condition is met more easily the higher is $\frac{\partial x(q_m^*,\overline{P})}{\partial q}$ in absolute value, so it is met more easily when $R_a^\theta(W_1^{*\theta}) - R_a^\theta(W_2^{*\theta})$ is larger, i.e., the more increasing (or less decreasing) is consumers' risk aversion.■

Proof of Proposition 2

At $p+\Delta p = \overline{P}$, the second term in (9) vanishes, so we are left with:

$$\frac{\partial \Pi^m(x_m^*, \overline{P})}{\partial \Delta p} = [q(x_m^*, \overline{P}) - 1]Hx_m^* - K \tag{24}$$

The industry's first order condition with respect to x yields:

$$[q(x_m^*, p + \Delta p) - 1] = -\frac{\partial q(x_m^*, p + \Delta p)}{\partial x} x_m^* \tag{25}$$

Substituting (25) into (24) we get:

$$\frac{\partial \Pi^m(x_m^*, \overline{P})}{\partial \Delta p} = -\frac{\partial q(x_m^*, \overline{P})}{\partial x}(x_m^*)^2 H - K$$

So the necessary and sufficient condition for the industry to set Δp such that $p+\Delta p > \overline{P}$ is:

$$-\frac{\partial q(x_m^*, \overline{P})}{\partial x}(x_m^*)^2 > \frac{K}{H} \tag{26}$$

To derive the condition for a duopolist to set $p + \Delta p^{*c} < \overline{P}$, for $p + \Delta p^{*c} = \overline{P}$, (10) turns into:

$$x_i^{*c} H \cdot [q(x_1^{*c} + x_2^{*c}, \overline{P}) - 1] + x_i^{*c}\overline{P}H\frac{\partial q(x_1^{*c} + x_2^{*c}, \overline{P})}{\partial(x_1 + x_2)}\frac{\partial x_j(\overline{P})}{\partial \Delta p} - K \tag{27}$$

Firm i's first order condition with respect to x_i is:

$$\frac{\partial q(x_1 + x_2, p + \Delta p)}{\partial x_i} H \cdot (p + \Delta p)x_i + [q(x_1 + x_2, p + \Delta p) - 1]H \cdot (p + \Delta p) = 0 \tag{28}$$

Which can be simplified to:

$$[q(x_1^{*c} + x_2^{*c}, p + \Delta p) - 1] = -\frac{\partial q(x_1^{*c} + x_2^{*c}, p + \Delta p)}{\partial x_i} x_i^{*c} \tag{29}$$

Substituting (29) into (27) yields that a duopolist sets $p + \Delta p^{*c} < \overline{P}$ if and only if:

$$-\frac{\partial q(x_1^{*c} + x_2^{*c}, \overline{P})}{\partial x_i} x_i^{*c}[x_i^{*c} - \overline{P}\frac{\partial x_j(\overline{P})}{\partial \Delta p}] < \frac{K}{H} \tag{30}$$

At $p + \Delta p = \overline{P}$, by (12), the sign of $\frac{\partial x_j(\overline{P})}{\partial \Delta p}$ is the same as the sign of $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},\overline{P})}{\partial x_i \partial \Delta p}$, so for any $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},\overline{P})}{\partial x_i \partial \Delta p} \geq 0$, we can eliminate the term $-\overline{P}\frac{\partial x_j(\overline{P})}{\partial \Delta p}$ in (30), so the condition

for a duopolist to increase risk in a way that increases demand ($p + \Delta p < \overline{P}$) while a collusive industry increases risk in a way that reduces demand ($p + \Delta p > \overline{P}$) is:

$$-\frac{\partial q(x_m^*, \overline{P})}{\partial x}(x_m^*)^2 > \frac{K}{H} > -\frac{\partial q(x_1^{*c} + x_2^{*c}, \overline{P})}{\partial x_i}(x_i^{*c})^2 \tag{31}$$

The right-hand-side of (31) is smaller than its left-hand side because:

$$-\frac{\partial q(x_1^{*c} + x_2^{*c}, \overline{P})}{\partial x_i}x_i^{*c} = [q(x_1^{*c} + x_2^{*c}, \overline{P}) - 1] < [q(x_m^*, \overline{P}) - 1] = -\frac{\partial q(x_m^*, \overline{P})}{\partial x}x_m^*$$

Where the first equality follows from firm i's first order condition with respect to x_i in (29), the next inequality follows due to the market power utilization effect, and the last equality follows from the collusive industry's first order condition in (25).

Hence, when the condition in the proposition is met, a collusive industry sets $p+\Delta p > \overline{P}$ while a duopolist sets $p + \Delta p < \overline{P}$ and consequently the duopolist always sets Δp^{*c} below the Δp_m^* that a collusive industry would set.

Proof of Corollary 2:

Since $\frac{K}{H}$ is small by the condition in the Corollary, both a collusive industry and a duopoly increase P in a way that decreases demand. This follows from the reasoning in the proof of Proposition 2, according to which the condition for this is:

$$[q(x_m^*, \overline{P}) - 1]x_m^* > [q(x_1^{*c} + x_2^{*c}, \overline{P}) - 1]x_i^{*c} > \frac{K}{H} \tag{32}$$

For any $p + \Delta p$, the quantity effect is manifested in the difference:

$$\left|\frac{\partial q(x_m^*, p+\Delta p)}{\partial \Delta p}\right| x_m^* \cdot (p + \Delta p)H - \left|\frac{\partial q(x_1^{*c}+x_2^{*c}, p+\Delta p)}{\partial \Delta p}\right| x_i^{*c} \cdot (p + \Delta p)H \tag{33}$$

The market power utilization effect is manifested in:

$$x_m^* H \cdot [q(x_m^*, p + \Delta p) - 1] - x_i^{*c} H \cdot [q(x_1^{*c} + x_2^{*c}, p + \Delta p) - 1] \tag{34}$$

For a parallel shift in the demand function (i.e., $\frac{\partial^2 q(x_1^{*c}+x_2^{*c}, p+\Delta p)}{\partial x \partial \Delta p} = 0$ for $\Delta p \in (0, \overline{\Delta p})$), the strategic effect is positive by (12), and the fact that demand is decreasing with P. Hence a sufficient condition for a duopolist to raise P more than does a collusive

industry is that the quantity effect be larger than the market power utilization effect:

$$\begin{aligned}&\left|\frac{\partial q(x_m^*,p+\Delta p)}{\partial \Delta p}\right| x_m^* \cdot (p+\Delta p)H - \left|\frac{\partial q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial \Delta p}\right| x_i^{*c} \cdot (p+\Delta p)H \\ &> x_m^* H \cdot [q(x_m^*, p+\Delta p) - 1] - x_i^{*c} H \cdot [q(x_1^{*c}+x_2^{*c}, p+\Delta p) - 1]\end{aligned} \tag{35}$$

We divide both sides by H, evaluate the inequation at $\Delta p = \Delta p^{*m}$, and use a collusive industry's first order condition with respect to Δp to rewrite (35) as:

$$\begin{aligned}&(p+\Delta p^{*m}) \left|\frac{\partial q(x_m^*,p+\Delta p^{*m})}{\partial \Delta p}\right| x_m^* \cdot (x_m^* - x_i^{*c}) \\ &> \overline{P} \left|\frac{\partial q(x_m^*,p+\Delta p^{*m})}{\partial \Delta p}\right| x_m^* \cdot (x_m^* - x_i^{*c}) \\ &> \left|\frac{\partial q(x_m^*,p+\Delta p^{*m})}{\partial \Delta p}\right| (p+\Delta p^{*m}) x_m^* - x_i^{*c} \cdot [q(x_1^{*c}+x_2^{*c}, p+\Delta p^{*m}) - 1] \\ &+ \frac{K}{H}\end{aligned} \tag{36}$$

Where the first line of (36) follows because $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x \partial \Delta p} = 0$ for $\Delta p \in (0, \overline{\Delta p})$, the second line stems from $p + \Delta p^{*m} > \overline{P}$, and in the third line, we replaced $x_m^* H \cdot [q(x_m^*, p + \Delta p) - 1]$ (from the second line of (35)) with $\left|\frac{\partial q(x_m^*,p+\Delta p^{*m})}{\partial \Delta p}\right| (p+\Delta p^{*m}) x_m^* + \frac{K}{H}$ (using the collusive industry's first order condition with respect to Δp following (9). Hence, a sufficient condition for the quantity effect to dominate the market power effect is:

$$\overline{P} > \frac{\left|\frac{\partial q(x_m^*,p+\Delta p^{*m})}{\partial \Delta p}\right| (p+\Delta p^{*m}) x_m^* - x_i^{*c} \cdot [q(x_1^{*c}+x_2^{*c}, p+\Delta p^{*m}) - 1] + \frac{K}{H}}{\left|\frac{\partial q(x_m^*,p+\Delta p^{*m})}{\partial \Delta p}\right| x_m^* \cdot (x_m^* - x_i^{*c})} \equiv \widehat{P} \tag{37}$$

Clearly, the strategic effect remains positive and the aggregate demand effect remains small for small enough $\frac{\partial^2 q(x_1^{*c}+x_2^{*c},p+\Delta p)}{\partial x \partial \Delta p}$ and $\frac{\partial^2 q(x_m^*,p+\Delta p)}{\partial x \partial \Delta p}$ in the relevant domain, so the result carries over to such cases. Under the condition in (37), at the point Δp_m^*, (9) equals zero while (10) is positive, and consequently $\Delta p^{*c} > \Delta p_m^*$.■

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